

## Article

# Evaluation of Artificial Neural Networks with Satellite Data Inputs for Daily, Monthly, and Yearly Solar Irradiation Prediction for Pakistan

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**Abstract:** Solar irradiation is the most critical parameter to consider when designing solar energy systems. The high cost and difficulty of measuring solar irradiation makes it impractical in every location. This study's primary objective was to develop an artificial neural network (ANN) model for global horizontal irradiation (GHI) prediction using satellite data inputs. Three types of ANN, namely, the feed forward neural network (FFNN), cascaded forward neural network (CFNN), and Elman neural network (EMNN), were tested. The findings revealed that altitude, relative humidity, and satellite GHI are the most effective parameters, as they are present in all the best-performing models. The best model for daily GHI prediction was FFNN, which decreased daily MAPE, RMSE, and MBE by 25.4%, 0.11 kWh/m<sup>2</sup>/d, and 0.01 kWh/m<sup>2</sup>/d. The FFNN daily MAPE, RMSE, and MBE values were 7.83%, 0.49 kWh/m<sup>2</sup>/d, and 0.01 kWh/m<sup>2</sup>/d. The EMNN performed best for monthly and annual prediction, reducing monthly MAPE, RMSE, and MBE by 50.62%, 0.13 kWh/m<sup>2</sup>/d, and 0.13 kWh/m<sup>2</sup>/d, while the reduction for yearly was 91.6%, 0.11 kWh/m<sup>2</sup>/d, 0.2 kWh/m<sup>2</sup>/d. The EMNN monthly MAPE, RMSE, and MBE values were 3.36%, 0.16 kWh/m<sup>2</sup>/d, and 0.16 kWh/m<sup>2</sup>/d, while the yearly values were 0.47%, 0.18 kWh/m<sup>2</sup>/d, and 0.004 kWh/m<sup>2</sup>/d.

**Keywords:** solar radiation; artificial neural network; prediction; satellite inputs; Pakistan



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## 1. Introduction

Due to growing concerns about climate change worldwide, the use of renewable energy resources, particularly solar energy, has increased during the last decade. These system's production cannot be planned according to demand because solar energy generation is dependent on uncontrollable weather and climatic variables. The intermittent nature of solar energy resources is a fundamental constraint to the greater use of renewable energy in the central power system [1]. Due to the random nature of solar power, even isolated solar power systems find it challenging to generate a constant output [2]. The shifting trends of renewable energy resources, particularly solar energy, should be assessed in order to be utilized [3]. Solar irradiation data can determine the shifting trends of solar energy. These data show how much energy a specific area receives from the sun at different times of the day, month, or year. This information is critical in the planning of solar energy systems. Physically measuring these data is nearly impossible due to the cost and challenges associated, especially in developing nations [4,5].

Researchers have developed numerous approaches to overcome the problems and expenses associated with evaluating global horizontal irradiation (GHI) on the ground. The best-studied models are empirical, physical, and artificial intelligence (AI)-based models. Researchers have used empirical models for GHI prediction and have shown excellent results. They can be classified into sunshine-based, modified, and non-sunshine-based models. Sunshine-based models only use the sunshine duration fraction as an input for GHI prediction, while modified sunshine-based models use other meteorological parameters, such as temperature and humidity, in addition to the sunshine duration fraction, and non-sunshine-based models do not use the sunshine fraction as an input, instead using other meteorological parameters for GHI prediction. Among these three models, the modified sunshine-based models predict more accurately, but the unavailability of sunshine hours data limits their use to areas in which meteorological stations record sunshine hours data [6–9]. Furthermore, due to dependence on extraterrestrial irradiation, the models are location-based, and each location-specific set of constants should be determined before using a model for GHI prediction [10]. L. Wang et al. [11] confirmed that the same empirical model accuracy changes with station location.

Physical models use the radiative transferring process for the estimation of solar irradiation. This method has accurately predicted solar radiation in several world regions [12]. These models are too complex, and numerous meteorological data, such as cloud cover, height, ozone, aerosol, and cloud type, are required to build a physical model. These data are generally not readily available, and if they are available, they make the model very complex. Due to their complexity and a lack of data, these models are rarely used [13].

In recent years, AI models have shown encouraging results in forecasting GHI [14]. With promising results, ANNs have recently been widely used in various parts of the world, including Turkey, Oman, China, India, Australia, and many other places [15].

For the prediction of average monthly solar radiation in Makurdi, Nigeria, Kuhe, Achirgbenda, and Agada [16] used a feed forward neural network (FFNN), a radial basis function network (RBFN), and a generalized regression neural network (GRNN). They used datasets from Makurdi meteorological stations to train and test the ANN model. The inputs used for the training of neural networks were relative humidity (RH), maximum temperature ( $T_{\max}$ ), minimum temperature ( $T_{\min}$ ), wind speed (WS), and sunshine hours (SH).

The hourly GHI forecasts of FFNNs, k-nearest neighbor (k-NN), auto regressive integrated moving average (ARIMA), and SVM, for the Tetouan region in Morocco were compared by Brahim Belmahdi, Mohamed Louzazni, and Abdelmajid El Bouardi [17]. Clearness Index ( $K_T$ ), maximum and minimum temperature difference ( $\Delta T$ ), ratio temperature, mean temperature ( $T_{\text{mean}}$ ),  $T_{\max}$ , and extraterrestrial irradiation ( $H_0$ ) were the input parameters used, which were determined using the Pearson coefficient test. For network performance evaluation, six distinct statistical metrics were used: root mean square error (RMSE), normalized root mean square error (NRMSE), mean absolute percentage error (MAPE), mean bias error (MBE), t-statistics (t-stat), and standard deviation ( $\sigma$ ). The statistical measures indicated that FFNNs trained with the LM approach outperform other models in terms of accuracy. The top performing model's MAPE, NRMSE, RMSE, MBE, and t-stat values were 1.80 percent, 0.57, 15.8, 23.88, and 6.68 percent.

Tegenu Argaw Woldegiyorgis et. Al. [18] compared the performance of their ANN model to three empirical models for Lalibela in Ethiopia. The empirical models included the Angstrom–Prescot model and the Louche model, which are sunshine duration-based models, and the third model was the Glover–McCulloch model, which, in addition to sunshine duration, also utilizes latitude as input. Three-year data were used for this study. MBE, RMSE, and  $R^2$  were used to evaluate the performance of each model. The ANN's  $R^2$  value was 0.799, while the empirical model values ranged from 0.126 to 0.17. The MBE values for the empirical models ranged from  $-0.468$  to  $-0.05$ , while for the ANN, MBE was  $-0.0005$ . Similarly, the RMSE value for the ANN was  $0.331 \text{ kWh/m}^2$ , while the RMSE values for the empirical models ranged from  $1.071 \text{ kWh/m}^2$  to  $1.263 \text{ kWh/m}^2$ . All three

statistical indicators confirmed that the ANN model's predictions were more accurate than the sunshine duration-based models.

Agbulut, Gurul, and Bicen [19] used four different machine learning techniques, FFNN, kernel nearest neighbor (k-NN), support vector machine (SVM), and deep learning, for daily GHI prediction in four Turkish cities. They used data from meteorological stations for  $T_{\max}$ ,  $T_{\min}$ , and cloud cover (CC), as well as two computed input factors, extraterrestrial radiation and SH. Compared to the other three methodologies, they found that the ANN model accurately forecasted solar radiation. For accuracy checking, they used seven different statistical metrics. Mahdy, Yousaf, and Dosky [20] investigated the effects of meteorological parameters on solar radiation. They developed seven distinct ANN models, each with its own set of input parameters. The input parameters were  $T_{\max}$ ,  $T_{\min}$ , CC, RH, WS, atmospheric pressure (AP), and ultraviolet irradiation (UVI). They calculated the daily GHI for Duhok City, Iraq, and compared this to data collected from the Directorate of Meteorology and Seismology station in Duhok City.

Ghimire et al. [21] used four different machine learning approaches and two deterministic models to assess solar energy potential in Queensland, Australia; the results revealed that the ANN outperformed all the other models. They used 36 years of data, including ERA-interim reanalysis and scientific data for landowners.

Sozen et al. [22] employed an FFNN network with the following input parameters: latitude (L), longitude (Lon), altitude (Alt), mean SH, month (m), day (d), and mean temperature ( $T_{\text{mean}}$ ) for solar mapping in Turkey. They used data from 11 cities for training and six cities to test the FFNN. The trained and tested FFNN showed high accuracy in cities where sun radiation data were unavailable.

Ozen Senkal and Kuleli [23] compared physical method results with ANN results in Turkey and discovered that the ANN results were considered superior to the physical method results. They used data from nine Turkish meteorological sites to train the ANN and data from three different locations to test it. They utilized L, Lon, Alt, mean diffuse radiation, and mean beam radiation for ANN training.

Khatib et al. [24] investigated the accuracy of linear, nonlinear, fuzzy, and ANN prediction accuracy for GHI predictions for five sites in Malaysia and discovered that ANN prediction accuracy was the highest of all the approaches. The mean absolute percentage error (MAPE) values for linear, nonlinear, fuzzy, and ANN were 8.13 percent, 6.93 percent, 6.71 percent, and 5.38 percent, respectively.

In 1957, the Pakistan Meteorological Department (PMD) started measuring solar radiation at its five stations in Islamabad, Karachi, Peshawar, Lahore, and Quetta, ceasing in 2000. The PMD also recorded sunshine duration from its 37 meteorological stations. Significant research on the solar mapping of Pakistan was conducted between 1986 and 1996; after 1996, no significant attempts were made to predict the GHI until 2007 [25]. In 2007, the National Renewable Energy Laboratory (NREL), USA, performed a satellite-based solar mapping of Pakistan. In 2015, the Energy Sector Management Assistance Program (ESMAP) of the World Bank started recording GHI for nine sites in Pakistan [26].

Sukhera and Pasha created the first model for solar radiation prediction using the Angström model in Pakistan in 1987 [27]. Table 1 lists other researchers who have created models based on empirical methods. Monthly and annual solar radiation maps show the average GHI data for each month and year. Daily maps show the same GHI received on a given day. They created an annual and monthly map for 58 locations in Pakistan using data from 37 stations for SH and 21 sites for  $T_{\max}$  and  $T_{\min}$ . To date, no attempt has been made in Pakistan to predict GHI using an ANN [25].

**Table 1.** Pakistan Solar Maps.

| Sr. No. | Ref.                          | Map Type           | Solar Radiation Data Used (No. of Stations) | Sunshine Duration Data Used (No. of Stations)      | Data Predicted for No. of Stations |
|---------|-------------------------------|--------------------|---------------------------------------------|----------------------------------------------------|------------------------------------|
| 1.      | Adnan et al. (2012) [28]      | Annual             | -                                           | 58 (37 for SH. 21 for $T_{\max}$ and $T_{\min}$ ). | 58                                 |
| 2.      | Ghaffar (1995) [29]           | Monthly            | 5                                           | 37                                                 | 37                                 |
| 3.      | Chaudry (1990) [30]           | Annual<br>Monthly  | 5                                           | 37                                                 | 37                                 |
| 4.      | Raja And Tidwell (1989) [31]  | Annual<br>Monthly  | 7 (2 sites from India)                      | 40                                                 | 40                                 |
| 5.      | Sukhera and Pasha (1987) [27] | Annual,<br>Monthly | 5                                           | 35                                                 | 35                                 |

ANNs have shown excellent performance in solar radiation prediction, as seen in [15,16,19–24]. Only FFNN, LSTM, RBFN, k-NN, and SVM performances have been tested in earlier studies, with FFNN providing the best results. There have been no studies on the accuracy of the cascaded forward neural network (CFNN) or Elman neural network (EMNN), the closest in operation to FFNN, for GHI prediction in the literature. The models developed employ input parameters recorded by meteorological stations, and they are relevant for regions with meteorological stations. On the other hand, meteorological stations are only present in major cities in developing countries such as Pakistan. It should be clarified that, for Pakistan, only empirical models for solar radiation prediction have been created; no AI-based model has been developed for Pakistan [25–32].

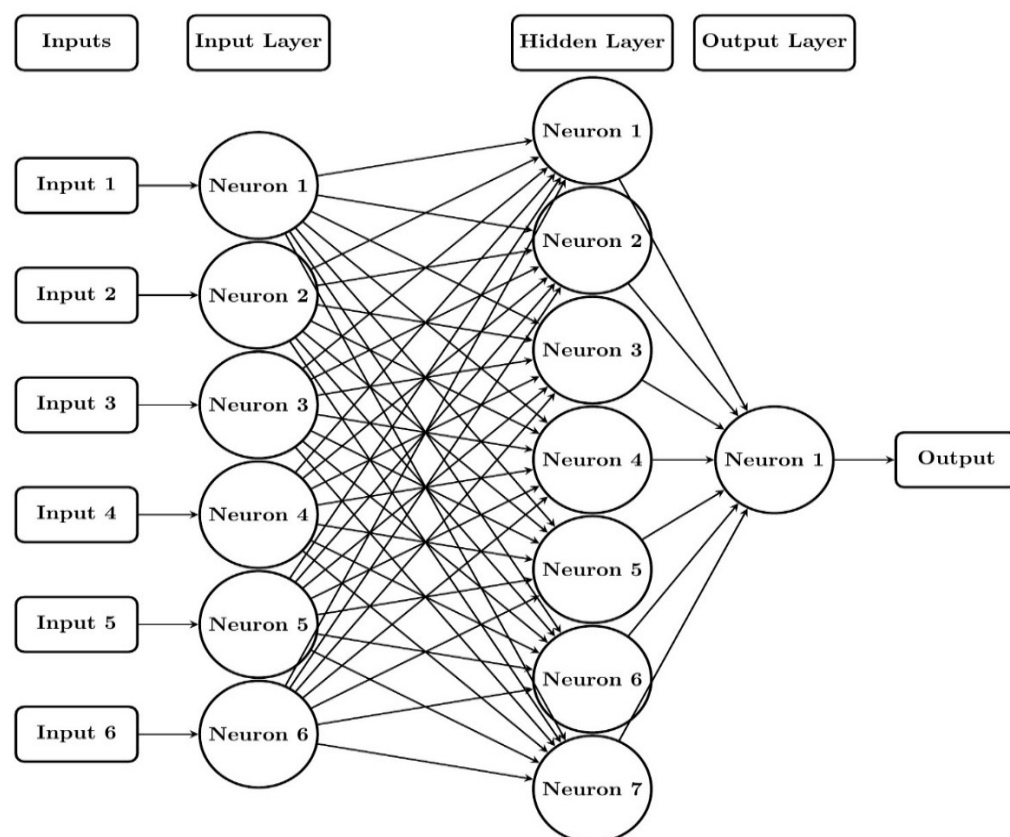
The only data available for all of Pakistan are NASA data, which overestimate sun irradiation by 25% in some locations. This information is used by solar energy planners to develop residential and commercial solar power projects. The use of NASA satellite data leads to project under-sizing, making plant operation unsustainable and putting project owners at risk of frequent power outages. This is one of the most significant obstacles to the sustainable development of solar energy in Pakistan.

This study aims to develop a more accurate GHI prediction model by correlating satellite data parameters and ground-measured GHI in order to make solar project operation and development more sustainable. We will evaluate the performance of feed forward neural networks (FFNNs), cascaded forward neural networks (CFNNs), and Elman neural networks (EMNNs) to achieve this goal. Satellite data are available for all places in Pakistan; however, they exaggerate GHI by up to 25% in some areas.

## 2. Artificial Neural Networks

An artificial neural network (ANN) is a human brain replica that connects multiple parameters in large amounts of nonlinear and uncertain data. The use of ANNs to solve problems does not necessitate any prior knowledge of complex mathematics. This means that ANNs take less computing effort to link input parameters to targets than other techniques [32]. Nonalgorithmic and nonlinear parallel information processing approaches are known as ANNs [4]. A neuron is the basic unit of an ANN, stacked in layers. Although an ANN can have many layers, they are grouped into input, hidden, and output layers. The input layer receives all information, the data processing begins in the hidden layer, and the results are determined in the output layer [33]. The input signal is multiplied by a weight, and then bias is added to it. The input layer signal is transmitted to the hidden layer, consisting of its activation function. The hidden layer processes this information and then passes it to the output layer, which has its activation function. Purelin, which has the same output format as input, is a typical activation function for output neurons [34]. In this study, we compared the performance of three ANN types for GHI prediction: FFNN, CFNN, EMNN. The FFNN model has demonstrated promising GHI prediction results,

but the other two networks have not been evaluated for daily, monthly, or annual GHI prediction. Figure 1 depicts the structure of an FFNN.



**Figure 1.** Feed forward neural network.

### 2.1. Feed Forward Neural Network (FFNN)

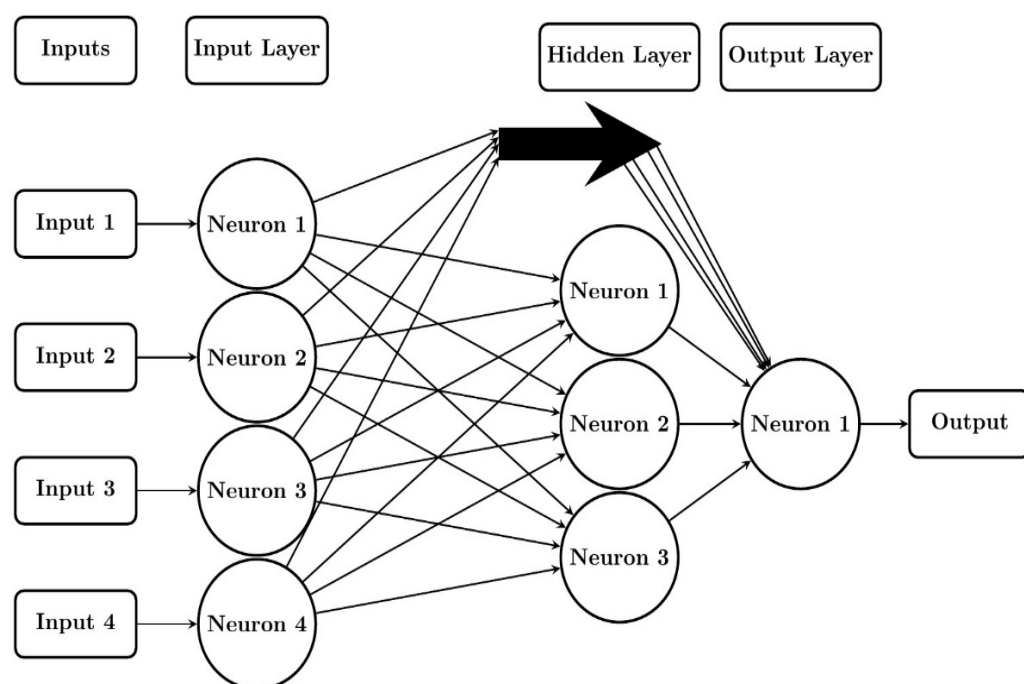
An FFNN is a multilayer perception (MLP) network with an input layer that receives inputs. The input layer standardizes the inputs, ensuring that each value falls between  $-1$  and  $1$ . The standardized inputs are then distributed to the hidden layer by the input layer. Standardized inputs are multiplied by weights, bias is introduced, and the resultant values are aggregated in the hidden layer. The output layer receives the combined value after passing through the transfer function. The result of the hidden layer is multiplied by weights in the output layer, and biases are then added to the output. The output layer transfer function receives the result from the output layer and produces the final projected results [35]. FFNNs have been extensively utilized by researchers in diverse domains with encouraging results [36]. They have been routinely used to predict GHI across the world.

### 2.2. Cascaded Forward Neural Network (CFNN)

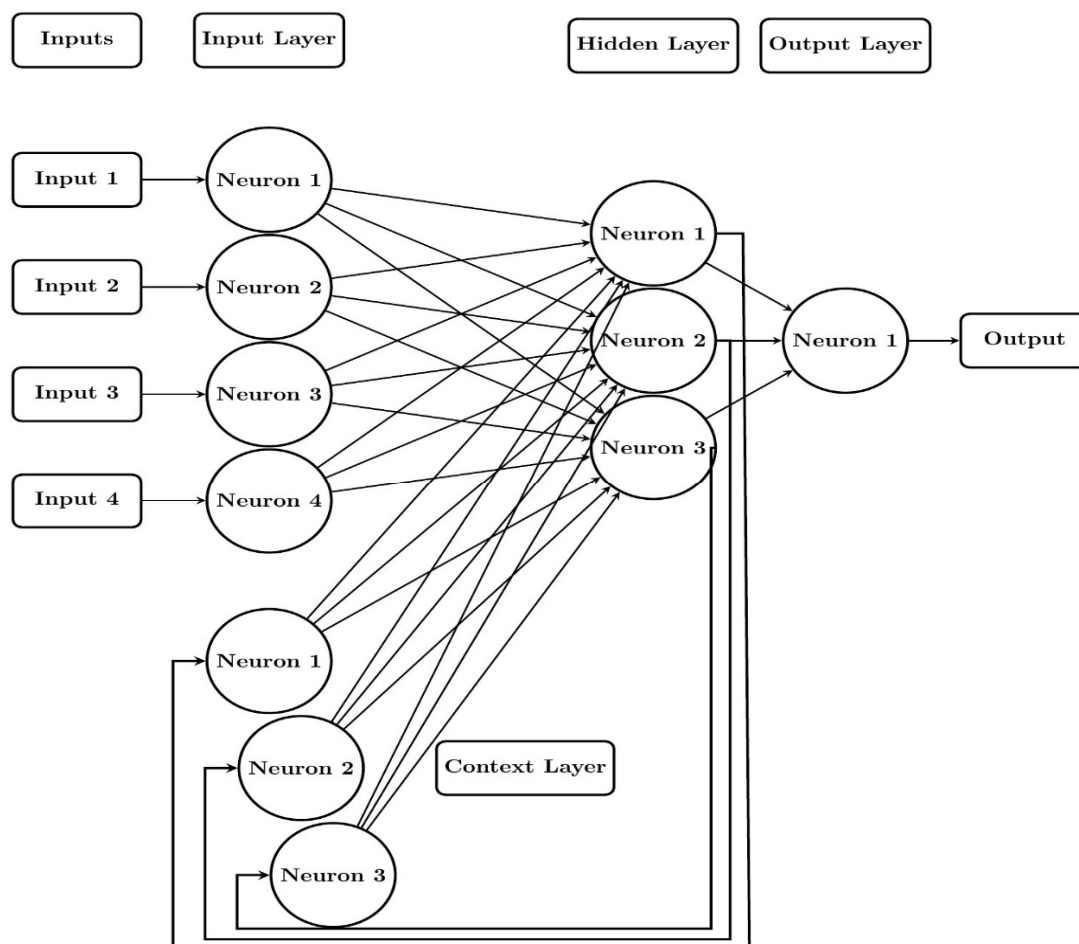
CFNNs also consist of input, hidden, and output layers in which neurons are arranged. In terms of operation, CFNNs are comparable to FFNNs. A forward pass across the network is also part of CFNN operation, with additional connections from the input to each layer and from each layer to successive layers [37]. Figure 2 shows the CFNN model.

### 2.3. Elman Neural Network (EFNN)

The main configuration of EMNNs is identical to that of FFNNs, but it incorporates an extra layer known as the context layer. The hidden layer neuron's output is sent to context layer neurons, which store the values. The stored values are fed back into the hidden layer to improve the correlation [37]. Figure 3 shows the EMNN model. The layers and connections of FFNN, CFNN, and EMNN are shown in Table 2.



**Figure 2.** Cascaded forward neural network.



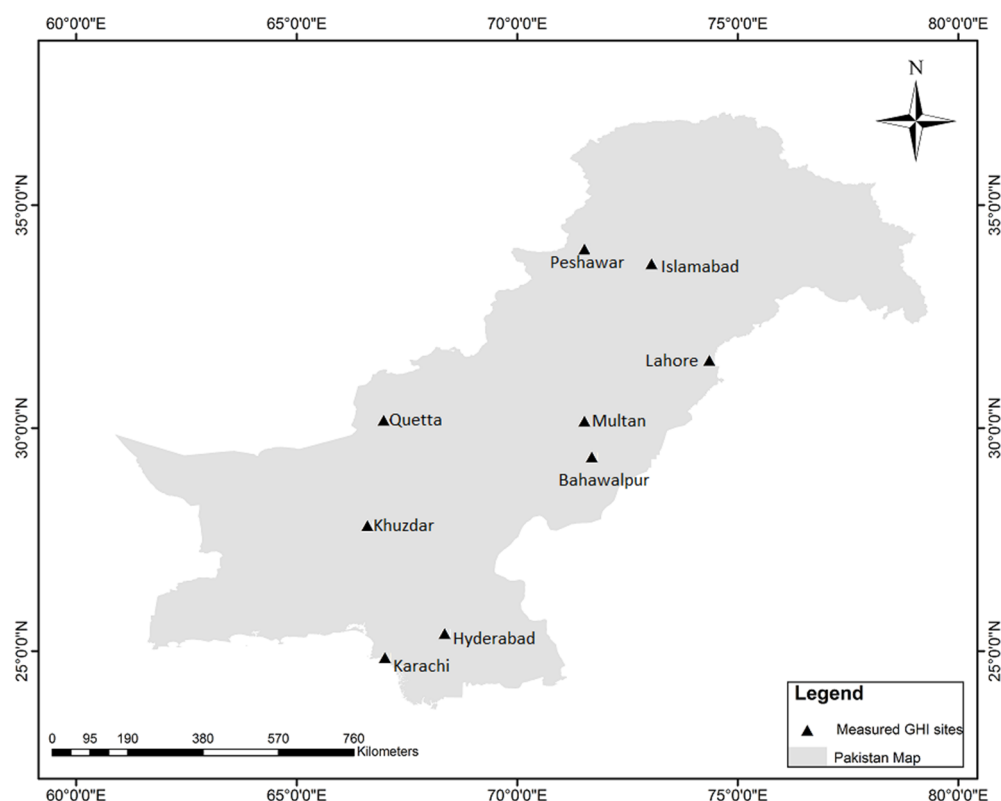
**Figure 3.** Elman neural network.

**Table 2.** ANN networks comparison.

| Network | Types of Layers                    | Connections                                                                   |
|---------|------------------------------------|-------------------------------------------------------------------------------|
| FFNN    | 3 (Input, Hidden, Output)          | 1. Inputs → Hidden → Output                                                   |
| CFNN    | 3 (Input, Hidden, Output)          | 1. Inputs → Hidden → Output<br>2. Inputs → Outputs                            |
| EMNN    | 4 (Input, Hidden, Output, Context) | 1. Inputs → Hidden → Output<br>2. Inputs → Hidden → Context → Hidden → Output |

### 3. Test Area and Data

Figure 4 depicts the nine locations (Bahawalpur, Hyderabad, Islamabad, Karachi, Khuzdar, Multan, Lahore, Peshawar, and Quetta) where ESMAP has been recently used to measure GHI data. Peshawar is the capital of Khyber Pakhtunkhwa, a province in Pakistan's northwest. Islamabad is Pakistan's capital city, located east of Peshawar. Lahore, Multan, and Bahawalpur are in the Punjab province, whereas Karachi and Hyderabad are located in the southern province of Sindh. Quetta is in the province of Balochistan. These location data were used to appropriately evaluate the FFNN, CFNN, and EMNN models. The locations are spread across the country's climatic and geographical zones.

**Figure 4.** ESMAP GHI-measuring sites.

Ground-measured GHI data were employed as targets in this study, with satellite data characteristics as inputs for the training and testing of ANNs. The ESMAP measured GHI for two years at a time interval of 10 min. The satellite data were downloaded from the NASA Power Data Viewer.

## 4. Methodology

### 4.1. Data Collection and Processing

All nine locations' data were downloaded from the ESMAP website. The data were collected at 10-min intervals. They were transformed into hourly, daily, and monthly formats. The data were cleaned before conversion by deleting any missing points. MS Excel was used to convert the data to hourly, daily, and monthly formats after they had been cleaned. The satellite data were downloaded in an hourly format from the NASA Power Data Viewer [38] and then converted to daily and monthly formats.

Following data organization, the data were normalized to a range of 0.1 to 0.9 for model consistency, using the equation as follows:

$$x_n = 0.8 \left( \frac{x - x_{min}}{x_{max} - x_{min}} \right) + 0.1 \quad (1)$$

where  $x_n$  is the normalized value and  $x$  is the original value, and  $x_{min}$  and  $x_{max}$  are the minimal and maximal values, respectively.

### 4.2. Development of ANN

For the development the ANN models, 12 meteorological and non-meteorological inputs (latitude (L), longitude (Lon), altitude (Alt), day (d), month (m), maximum temperature ( $T_{max}$ ), minimum temperature ( $T_{min}$ ), relative humidity (RH), absolute humidity (AH), Kt, and satellite GHI ( $G_{sat}$ )) were used as inputs, and 45 different combinations of these parameters were developed, resulting in 45 ANN models with different input combinations. The ANN models were developed using MATLAB version R2019b on a Core i7 10th generation laptop.

### 4.3. Training and Testing of ANN

The data from eight locations were used to train ANNs, while the ninth site's data were used to test the ANNs. This approach was repeated nine times to use all nine sites as test sites, resulting in nine datasets that were used to determine the accuracy of the ANNs across all areas. After the publication of ESMAP data in 2019, the data were downloaded, and ANN simulations using MATLAB began in March 2021.

The performance of FFNN, CFNN, and EMNN was assessed. As a training algorithm, Levenberg–Marquardt was utilized. In addition, 45 ANN models with subsets of the twelve input parameters, stated in Section 4.2, were generated and an FFNN was used to check the accuracy of all 45 models. For each model, up to 60 hidden layer neurons were examined to determine the number of neurons with the maximum accuracy. Then, based on accuracy, we chose the best eight models, as shown in Table 3, and we used these models to assess the performance of the CFNN and EMNN models.

**Table 3.** ANN models with input parameters.

| ANN Model | Inputs                                                              |
|-----------|---------------------------------------------------------------------|
| ANN10     | L, Lon, Alt, d, $T_{max}$ , $T_{min}$ , RH, AH, KT, Prec, $G_{sat}$ |
| ANN11     | L, Lon, Alt, d, $T_{max}$ , $T_{min}$ , RH, AH, $G_{sat}$           |
| ANN13     | L, Lon, Alt, m, d, $T_{max}$ , $T_{min}$ , RH, AH, $G_{sat}$        |
| ANN29     | L, Lon, Alt, m, d, Prec, RH, $G_{sat}$                              |
| ANN30     | L, Lon, Alt, d, Prec, RH, $G_{sat}$                                 |
| ANN31     | L, Lon, Alt, m, d, Prec, RH, AH, $G_{sat}$                          |
| ANN43     | L, Lon, Alt, d, $G_{sat}$                                           |
| ANN45     | Alt, m, d, $T_{max}$ , $T_{min}$ , RH, AH, $G_{sat}$                |

### 4.4. Performance Evaluation Using Statistical Metrics

After finishing the predictions for all models and all three networks, the predicted GHI data were compared with the measured GHI data using statistical metrics. For ANN

accuracy assessment, MAPE, RMSE, and MBE were used in this study. The equations for calculating these statistical metrics are provided in Equations (2)–(4) as follows:

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \frac{\text{GHI} - \text{GHI}_{\text{ANN}}}{\text{GHI}} \quad (2)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\text{GHI}_{\text{ANN}} - \text{GHI})^2} \quad (3)$$

$$\text{MBE} = \frac{1}{n} \sum_{i=1}^n (\text{GHI}_{\text{ANN}} - \text{GHI}) \quad (4)$$

where GHI represents the measured value,  $\text{GHI}_{\text{ANN}}$  means the predicted value, and  $n$  denotes the total number of datapoints tested [39]. The higher the precision, the lower the MAPE, RMSE, and MBE values are. MAPE was used to classify the ANN model. A MAPE of less than 10% implied great accuracy [40]. After that, the average MAPE of each model was calculated to determine the model that can be used for any location across the country.

## 5. Results and Discussion

Using satellite data inputs and the ESMAP GHI data as targets, twenty-four models (FFNN = 8, CFFN = 8, EMNN = 8, shown in Table 3) were trained nine times to verify the accuracy of each model for each location. The use of satellite data as inputs removes the limitation of previous ANN models that can only be used for sites where meteorological stations are available. Furthermore, the testing of different ANN networks for Pakistan proves that all the networks have the potential to predict GHI accurately. A comparison of the FFNN, CFNN, and EMNN models has shown that, as the time frame increases, the results of all networks improve. Secondly, the tick and trial method for neuron selection was very effective and helped select the most effective number of neurons. In addition, testing the different combinations of input parameters helped in finding the input parameters that influence the results. We predicted daily GHI for each site first; then monthly GHI for each site; and, lastly, calculated annual GHI for each site. Table 4 shows the daily, monthly, and yearly satellite data MAPE, RMSE, and MBE for each site and the average of all locations. Table 4 shows that the satellite data overpredict daily, monthly, and yearly data, with MAPE ranging from 5.53% in Hyderabad to 25.31% in Lahore, 1.78% in Quetta to 15.27 percent in Lahore, and 0.24 percent in Multan to 15.27 percent in Lahore. Daily GHI RMSE ranges from 0.41 kWh/m<sup>2</sup>/d in Multan to 1.07 kWh/m<sup>2</sup>/d in Lahore, monthly RMSE from 0.10 kWh/m<sup>2</sup>/d to 0.53 kWh/m<sup>2</sup>/d in Lahore, and annual RMSE from −0.10 kWh/m<sup>2</sup>/d in Quetta to 0.29 kWh/m<sup>2</sup>/d in Lahore.

**Table 4.** Satellite data evaluation metrics.

| Site       | MAPE (%) | Daily                           |                                |             | Monthly                         |                                |             | Annual                          |                                |
|------------|----------|---------------------------------|--------------------------------|-------------|---------------------------------|--------------------------------|-------------|---------------------------------|--------------------------------|
|            |          | RMSE<br>(kWh/m <sup>2</sup> /d) | MBE<br>(kWh/m <sup>2</sup> /d) | MAPE<br>(%) | RMSE<br>(kWh/m <sup>2</sup> /d) | MBE<br>(kWh/m <sup>2</sup> /d) | MAPE<br>(%) | RMSE<br>(kWh/m <sup>2</sup> /d) | MBE<br>(kWh/m <sup>2</sup> /d) |
| Lahore     | 25.31    | 1.07                            | 0.53                           | 15.27       | 0.53                            | 0.53                           | 5.25        | 0.29                            | 0.24                           |
| Peshawar   | 18.32    | 0.85                            | 0.48                           | 11.62       | 0.47                            | 0.47                           | 11.62       | 0.47                            | 0.47                           |
| Bahawalpur | 8.58     | 0.57                            | −0.30                          | 5.80        | 0.33                            | 0.33                           | 4.63        | 0.33                            | 0.30                           |
| Hyderabad  | 5.53     | 0.40                            | −0.26                          | 5.01        | 0.30                            | 0.30                           | 4.09        | 0.30                            | 0.26                           |
| Islamabad  | 9.00     | 0.47                            | 0.18                           | 4.38        | 0.18                            | 0.18                           | 4.36        | 0.18                            | 0.18                           |
| Karachi    | 6.93     | 0.43                            | 0.01                           | 3.44        | 0.17                            | 0.17                           | 1.01        | 0.17                            | 0.01                           |
| Khuzdar    | 7.82     | 0.67                            | −0.35                          | 5.59        | 0.35                            | 0.35                           | 5.59        | 0.35                            | 0.35                           |
| Multan     | 7.37     | 0.41                            | −0.08                          | 4.51        | 0.20                            | 0.20                           | 0.24        | 0.20                            | 0.07                           |
| Quetta     | 5.67     | 0.49                            | −0.03                          | 1.78        | 0.10                            | 0.10                           | 0.47        | 0.10                            | 0.03                           |
| Average    | 10.50    | 0.60                            | 0.02                           | 6.38        | 0.29                            | 0.29                           | 5.25        | 0.29                            | 0.24                           |

For satellite data, the average daily, monthly, and annual MAPE, RMSE, and MBE values were (10.50%, 0.60 kWh/m<sup>2</sup>/d, 0.02 kWh/m<sup>2</sup>/d), (6.38%, 0.29 kWh/m<sup>2</sup>/d, 0.29 kWh/m<sup>2</sup>/d), and (−1.92%, 0.29 kWh/m<sup>2</sup>/d).

Table 5 provides the daily, monthly, and annual evaluation metrics, including MAPE, RMSE, and MBE for the FFNN, CFNN, and EMNN models. It is clear from Tables 4 and 5 that the networks predicted GHI better in terms of MAPE, RMSE and MBE for daily, monthly, and annual data than satellites. Except for Lahore and Peshawar, all the models in Table 5 can be classified as high accuracy models [33].

**Table 5.** Evaluation of the best ANN models in each network.

| Site       | Network | Daily    |                              |                             | Monthly  |                              |                             | Annual   |                              |                             |
|------------|---------|----------|------------------------------|-----------------------------|----------|------------------------------|-----------------------------|----------|------------------------------|-----------------------------|
|            |         | MAPE (%) | RMSE (kWh/m <sup>2</sup> /d) | MBE (kWh/m <sup>2</sup> /d) | MAPE (%) | RMSE (kWh/m <sup>2</sup> /d) | MBE (kWh/m <sup>2</sup> /d) | MAPE (%) | RMSE (kWh/m <sup>2</sup> /d) | MBE (kWh/m <sup>2</sup> /d) |
| Lahore     | FFNN    | 16.49    | 0.77                         | 0.01                        | 8.38     | 0.33                         | 0.33                        | 0.89     | 0.33                         | 0.004                       |
|            | CFNN    | 18.95    | 0.85                         | 0.12                        | 8.48     | 0.35                         | 0.35                        | 0.37     | 0.37                         | 0.05                        |
|            | EMNN    | 19.54    | 0.89                         | 0.03                        | 9.00     | 0.34                         | 0.34                        | 0.69     | 0.43                         | 0.00                        |
| Peshawar   | FFNN    | 11.75    | 0.66                         | 0.09                        | 3.23     | 0.13                         | 0.13                        | 0.17     | 0.13                         | 0.03                        |
|            | CFNN    | 13.02    | 0.69                         | 0.04                        | 3.67     | 0.17                         | 0.17                        | 0.74     | 0.21                         | 0.08                        |
|            | EMNN    | 12.44    | 0.72                         | 0.22                        | 3.70     | 0.14                         | 0.14                        | 0.15     | 0.16                         | 0.00                        |
| Bahawalpur | FFNN    | 6.36     | 0.44                         | 0.01                        | 2.57     | 0.11                         | 0.11                        | 0.83     | 0.16                         | 0.01                        |
|            | CFNN    | 6.37     | 0.43                         | 0.09                        | 2.02     | 0.11                         | 0.11                        | 0.29     | 0.12                         | 0.03                        |
|            | EMNN    | 6.68     | 0.46                         | 0.06                        | 2.26     | 0.13                         | 0.13                        | 0.09     | 0.16                         | 0.04                        |
| Hyderabad  | FFNN    | 3.37     | 0.24                         | 0.03                        | 1.36     | 0.08                         | 0.08                        | 0.13     | 0.12                         | 0.02                        |
|            | CFNN    | 3.58     | 0.26                         | 0.05                        | 2.01     | 0.11                         | 0.11                        | 0.15     | 0.11                         | 0.01                        |
|            | EMNN    | 3.56     | 0.25                         | 0.00                        | 2.00     | 0.11                         | 0.11                        | 0.05     | 0.11                         | 0.004                       |
| Islamabad  | FFNN    | 8.07     | 0.46                         | 0.06                        | 2.34     | 0.12                         | 0.12                        | 0.65     | 0.14                         | 0.06                        |
|            | CFNN    | 7.79     | 0.45                         | 0.05                        | 2.73     | 0.13                         | 0.13                        | 0.04     | 0.16                         | 0.02                        |
|            | EMNN    | 7.92     | 0.45                         | 0.02                        | 2.74     | 0.13                         | 0.13                        | 0.13     | 0.13                         | 0.04                        |
| Karachi    | FFNN    | 6.28     | 0.41                         | −0.06                       | 2.74     | 0.15                         | 0.15                        | 0.01     | 0.17                         | −0.03                       |
|            | CFNN    | 6.17     | 0.40                         | 0.04                        | 1.93     | 0.09                         | 0.09                        | 0.38     | 0.15                         | −0.01                       |
|            | EMNN    | 6.40     | 0.41                         | −0.06                       | 2.46     | 0.11                         | 0.11                        | 0.02     | 0.18                         | −0.03                       |
| Khuzdar    | FFNN    | 5.59     | 0.56                         | −0.06                       | 1.56     | 0.10                         | 0.10                        | 0.20     | 0.10                         | −0.02                       |
|            | CFNN    | 5.59     | 0.56                         | −0.06                       | 1.56     | 0.10                         | 0.10                        | 0.20     | 0.10                         | −0.02                       |
|            | EMNN    | 5.55     | 0.56                         | −0.07                       | 1.74     | 0.11                         | 0.11                        | −0.05    | 0.12                         | −0.01                       |
| Multan     | FFNN    | 6.07     | 0.37                         | −0.04                       | 1.93     | 0.10                         | 0.10                        | 0.25     | 0.10                         | −0.0002                     |
|            | CFNN    | 5.76     | 0.35                         | 0.0001                      | 1.82     | 0.08                         | 0.08                        | 0.01     | 0.10                         | −0.0001                     |
|            | EMNN    | 6.05     | 0.35                         | −0.003                      | 1.82     | 0.09                         | 0.09                        | 0.34     | 0.10                         | −0.01                       |
| Quetta     | FFNN    | 5.25     | 0.47                         | 0.02                        | 1.65     | 0.09                         | 0.09                        | 0.07     | 0.13                         | 0.01                        |
|            | CFNN    | 5.41     | 0.47                         | 0.002                       | 1.81     | 0.10                         | 0.10                        | 0.10     | 0.14                         | 0.02                        |
|            | EMNN    | 5.35     | 0.47                         | 0.02                        | 1.67     | 0.10                         | 0.10                        | 0.04     | 0.11                         | 0.01                        |
| Average    | FFNN    | 7.83     | 0.49                         | −0.01                       | 3.36     | 0.16                         | 0.16                        | 0.99     | 0.17                         | 0.05                        |
|            | CFNN    | 8.22     | 0.50                         | 0.003                       | 3.22     | 0.16                         | 0.16                        | 0.86     | 0.18                         | 0.06                        |
|            | EMNN    | 8.23     | 0.51                         | −0.05                       | 3.16     | 0.16                         | 0.16                        | 0.47     | 0.18                         | 0.04                        |

Tables 6–8 show each model's daily, monthly, and annual MAPE for each city and the average MAPE of each model for the whole country. Among the models depicted in Table 6, ANN10 had the best daily predictions out of all the networks. For monthly, as shown in Table 7, ANN45, ANN31, and ANN29 predicted best using the FFNN, CFNN, and EMNN models.

For annual predictions, ANN30 predicted best using the FFNN and EMNN models, while ANN43 predicted best using the CFNN model. The results of the ANN10 model had the highest accuracy for daily prediction among all the models. The best model for monthly prediction was ANN45, and the most accurate model for annual was ANN30.

The FFNN results were more accurate for daily predictions, as the MAPEs for daily GHI prediction ranged from 3.37% in Hyderabad to 16.49% in Lahore. The MAPE for the FFNN model was 7.83% on average. The FFNN RMSE for daily predictions varied from 0.24 kWh/m<sup>2</sup>/d in Hyderabad to 0.77 kWh/m<sup>2</sup>/d in Lahore. The FFNN model has an overall RMSE of 0.49 kWh/m<sup>2</sup>/d. The FFNN model's daily MBE was 0.01 kWh/m<sup>2</sup>/d, ranging from 0.000 kWh/m<sup>2</sup>/d in Multan to 0.22 kWh/m<sup>2</sup>/d in Peshawar. Monthly MAPE ranges from 1.36% in Hyderabad to 8.38% in Lahore, with RMSE and MBE values ranging from 0.08 kWh/m<sup>2</sup>/d in Multan to 0.33 kWh/m<sup>2</sup>/d in Lahore. For annual the GHI prediction, the FFNN model's average MAPE, RMSE, and MBE were 0.99%, 0.17 kWh/m<sup>2</sup>/d, and 0.05 kWh/m<sup>2</sup>/d.

**Table 6.** Daily MAPE values for each city and model.

| Cities     | ANN10 | ANN11 | ANN13 | ANN29 | ANN30 | ANN31 | ANN43 | ANN45 |
|------------|-------|-------|-------|-------|-------|-------|-------|-------|
| MAPE (%)   |       |       |       |       |       |       |       |       |
| FFNN       |       |       |       |       |       |       |       |       |
| Peshawar   | 11.75 | 12.66 | 12.58 | 12.33 | 13.08 | 12.47 | 12.7  | 12.67 |
| Bahawalpur | 6.35  | 6.80  | 6.85  | 6.98  | 7.18  | 7.15  | 7.35  | 6.69  |
| Hyderabad  | 3.50  | 3.48  | 3.78  | 3.82  | 3.75  | 3.37  | 3.95  | 3.67  |
| Islamabad  | 8.49  | 8.88  | 8.84  | 8.26  | 8.09  | 8.95  | 8.07  | 8.86  |
| Karachi    | 6.96  | 6.27  | 6.49  | 6.81  | 6.84  | 6.72  | 6.44  | 6.93  |
| Khuzdar    | 5.59  | 5.67  | 5.70  | 5.85  | 5.97  | 5.76  | 5.94  | 5.71  |
| Multan     | 6.07  | 6.42  | 6.63  | 6.70  | 6.85  | 6.49  | 7.00  | 6.52  |
| Quetta     | 5.25  | 5.45  | 5.39  | 5.61  | 5.66  | 5.40  | 5.77  | 5.63  |
| Lahore     | 16.48 | 20.24 | 19.99 | 19.78 | 20.07 | 20.18 | 20.19 | 23.83 |
| Average    | 7.83  | 8.43  | 8.47  | 8.46  | 8.61  | 8.50  | 8.60  | 8.95  |
| CFNN       |       |       |       |       |       |       |       |       |
| Peshawar   | 13.02 | 14.13 | 13.46 | 14.00 | 13.71 | 13.44 | 13.76 | 13.28 |
| Bahawalpur | 6.36  | 6.77  | 6.63  | 7.28  | 7.39  | 6.67  | 7.32  | 6.91  |
| Hyderabad  | 3.57  | 3.79  | 3.83  | 3.69  | 3.94  | 3.70  | 3.95  | 3.93  |
| Islamabad  | 8.97  | 8.78  | 8.89  | 8.21  | 7.93  | 8.77  | 7.79  | 8.76  |
| Karachi    | 6.29  | 6.65  | 6.17  | 6.31  | 7.07  | 6.49  | 6.36  | 6.56  |
| Khuzdar    | 5.59  | 5.67  | 5.70  | 5.85  | 5.97  | 5.76  | 5.94  | 5.71  |
| Multan     | 5.76  | 6.13  | 6.18  | 6.60  | 6.53  | 6.15  | 7.00  | 6.53  |
| Quetta     | 5.45  | 5.54  | 5.41  | 5.74  | 5.66  | 5.40  | 5.69  | 5.61  |
| Lahore     | 18.95 | 21.20 | 20.45 | 19.01 | 19.71 | 19.49 | 19.62 | 24.12 |
| Average    | 8.22  | 8.74  | 8.53  | 8.52  | 8.66  | 8.43  | 8.60  | 9.04  |
| EMNN       |       |       |       |       |       |       |       |       |
| Peshawar   | 12.44 | 13.06 | 13.72 | 13.29 | 13.20 | 12.78 | 13.50 | 12.70 |
| Bahawalpur | 6.67  | 7.02  | 6.84  | 7.04  | 7.13  | 7.16  | 7.32  | 6.79  |
| Hyderabad  | 3.56  | 3.63  | 3.98  | 3.96  | 3.94  | 3.70  | 3.95  | 3.93  |
| Islamabad  | 8.49  | 8.93  | 8.89  | 8.35  | 8.04  | 8.95  | 7.92  | 9.04  |
| Karachi    | 6.40  | 6.48  | 6.45  | 6.68  | 6.63  | 6.79  | 6.60  | 6.86  |
| Khuzdar    | 5.55  | 5.70  | 5.65  | 5.90  | 5.85  | 5.67  | 5.98  | 5.91  |
| Multan     | 6.04  | 6.46  | 6.59  | 6.29  | 6.62  | 6.51  | 6.99  | 6.06  |
| Quetta     | 5.34  | 5.45  | 5.34  | 5.68  | 5.77  | 5.46  | 5.81  | 5.76  |
| Lahore     | 19.54 | 20.03 | 21.82 | 20.46 | 20.03 | 20.45 | 20.71 | 23.56 |
| Average    | 8.23  | 8.52  | 8.81  | 8.63  | 8.58  | 8.61  | 8.75  | 8.96  |

**Table 7.** Monthly MAPE values for each city and model.

| Cities     | ANN10 | ANN11 | ANN13 | ANN29 | ANN30 | ANN31 | ANN43 | ANN45 |
|------------|-------|-------|-------|-------|-------|-------|-------|-------|
| MAPE (%)   |       |       |       |       |       |       |       |       |
| FFNN       |       |       |       |       |       |       |       |       |
| Peshawar   | 5.90  | 5.10  | 4.28  | 4.51  | 4.59  | 4.64  | 5.18  | 3.22  |
| Bahawalpur | 3.18  | 2.66  | 2.57  | 3.02  | 3.27  | 3.37  | 3.31  | 2.70  |
| Hyderabad  | 1.52  | 1.95  | 2.14  | 2.33  | 2.28  | 1.35  | 2.37  | 2.13  |
| Islamabad  | 4.01  | 4.52  | 3.87  | 3.05  | 2.33  | 4.05  | 2.78  | 3.97  |
| Karachi    | 4.07  | 2.73  | 2.82  | 3.58  | 3.45  | 3.01  | 3.32  | 3.45  |
| Khuzdar    | 2.41  | 1.84  | 1.81  | 1.61  | 2.40  | 1.72  | 2.11  | 1.55  |
| Multan     | 1.92  | 2.02  | 2.37  | 2.85  | 3.33  | 2.40  | 3.18  | 3.16  |
| Quetta     | 1.81  | 2.27  | 1.67  | 2.1   | 2.44  | 1.90  | 2.18  | 1.65  |
| Lahore     | 8.37  | 9.97  | 8.68  | 11.3  | 8.86  | 9.88  | 11.82 | 8.37  |
| Average    | 3.69  | 3.67  | 3.36  | 3.82  | 3.66  | 3.59  | 4.03  | 3.336 |

Table 7. Cont.

| Cities     | ANN10 | ANN11 | ANN13 | ANN29 | ANN30 | ANN31 | ANN43 | ANN45 |
|------------|-------|-------|-------|-------|-------|-------|-------|-------|
| CFNN       |       |       |       |       |       |       |       |       |
| Peshawar   | 6.30  | 6.30  | 3.89  | 4.92  | 4.79  | 5.31  | 5.04  | 3.67  |
| Bahawalpur | 2.02  | 2.61  | 2.54  | 3.43  | 3.66  | 2.20  | 3.51  | 3.22  |
| Hyderabad  | 2.23  | 2.13  | 2.50  | 2.09  | 2.29  | 2.00  | 2.37  | 2.10  |
| Islamabad  | 4.86  | 4.69  | 4.36  | 3.67  | 2.73  | 4.51  | 2.72  | 4.39  |
| Karachi    | 2.73  | 2.90  | 1.92  | 2.58  | 3.80  | 2.77  | 2.93  | 2.82  |
| Khuzdar    | 2.41  | 1.84  | 1.81  | 1.61  | 2.40  | 1.72  | 2.11  | 1.55  |
| Multan     | 2.04  | 2.01  | 2.05  | 2.99  | 3.29  | 1.81  | 3.18  | 1.94  |
| Quetta     | 1.81  | 2.30  | 1.92  | 2.52  | 2.44  | 1.90  | 2.00  | 1.94  |
| Lahore     | 10.86 | 11.12 | 11.39 | 8.48  | 9.95  | 8.47  | 10.54 | 15.63 |
| Average    | 3.92  | 3.99  | 3.60  | 3.59  | 3.93  | 3.41  | 3.82  | 4.14  |
| EMNN       |       |       |       |       |       |       |       |       |
| Peshawar   | 6.77  | 5.13  | 6.59  | 3.70  | 4.38  | 5.29  | 4.71  | 4.32  |
| Bahawalpur | 2.26  | 3.48  | 2.84  | 2.63  | 3.12  | 3.55  | 3.36  | 2.88  |
| Hyderabad  | 1.99  | 2.02  | 2.43  | 1.99  | 2.29  | 2.008 | 2.37  | 2.10  |
| Islamabad  | 4.35  | 4.62  | 4.48  | 3.55  | 2.74  | 4.63  | 2.72  | 4.48  |
| Karachi    | 2.63  | 2.62  | 2.46  | 3.36  | 3.51  | 3.18  | 3.41  | 3.57  |
| Khuzdar    | 2.21  | 1.86  | 1.90  | 1.92  | 2.12  | 1.74  | 2.05  | 1.99  |
| Multan     | 2.04  | 2.53  | 2.20  | 2.08  | 3.16  | 2.30  | 3.35  | 1.82  |
| Quetta     | 1.67  | 1.66  | 1.40  | 2.24  | 2.52  | 1.86  | 2.40  | 262   |
| Lahore     | 9.84  | 10.71 | 10.86 | 9.59  | 9.56  | 10.88 | 9.00  | 14.63 |
| Average    | 3.75  | 3.85  | 3.91  | 3.45  | 3.75  | 3.94  | 3.71  | 4.27  |

Table 8. Annual MAPE values for each city and model.

| Cities     | ANN10 | ANN11 | ANN13 | ANN29 | ANN30 | ANN31 | ANN43 | ANN45 |
|------------|-------|-------|-------|-------|-------|-------|-------|-------|
| MAPE (%)   |       |       |       |       |       |       |       |       |
| FFNN       |       |       |       |       |       |       |       |       |
| Peshawar   | 1.57  | 3.64  | 2.61  | 3.15  | 2.37  | 2.20  | 4.11  | 0.16  |
| Bahawalpur | 0.83  | 1.63  | 1.48  | 0.93  | 0.77  | 1.99  | 0.86  | 2.59  |
| Hyderabad  | 0.48  | 0.67  | 0.13  | 0.66  | 0.07  | 0.46  | 0.15  | 0.28  |
| Islamabad  | 1.12  | 2.01  | 0.71  | 1.23  | 0.88  | 2.57  | 0.65  | 3.19  |
| Karachi    | 3.9   | 0.53  | 0.19  | 1.55  | 0.85  | 2.16  | 0.36  | 0.01  |
| Khuzdar    | 0.66  | 1.05  | 0.76  | 0.20  | 1.15  | 0.00  | 0.74  | 0.81  |
| Multan     | 0.94  | 0.25  | 1.16  | 0.67  | 0.48  | 0.54  | 2.05  | 2.63  |
| Quetta     | 0.16  | 0.20  | 0.16  | 0.06  | 0.10  | 0.18  | 0.31  | 057   |
| Lahore     | 0.89  | 2.74  | 3.00  | 5.74  | 2.27  | 3.97  | 3.89  | 0.89  |
| Average    | 1.18  | 1.41  | 1.13  | 1.58  | 0.99  | 1.56  | 1.46  | 1.24  |
| CFNN       |       |       |       |       |       |       |       |       |
| Peshawar   | 1.39  | 1.39  | 1.30  | 2.95  | 0.74  | 4.41  | 1.18  | 2.40  |
| Bahawalpur | 1.55  | 1.33  | 1.53  | 0.85  | 0.86  | 0.28  | 1.09  | 2.64  |
| Hyderabad  | 0.53  | 0.14  | 1.87  | 0.70  | 0.22  | 0.35  | 0.15  | 0.32  |
| Islamabad  | 0.10  | 0.74  | 1.60  | 0.03  | 0.21  | 2.03  | 0.47  | 3.56  |
| Karachi    | 1.21  | 0.038 | 1.05  | 1.33  | 3.11  | 1.54  | 0.26  | 0.96  |
| Khuzdar    | 0.66  | 1.05  | 0.76  | 0.20  | 1.15  | 0.00  | 0.74  | 0.81  |
| Multan     | 0.01  | 0.89  | 0.39  | 0.80  | 1.52  | 0.21  | 2.05  | 1.21  |
| Quetta     | 0.38  | 1.68  | 1.07  | 0.98  | 0.10  | 0.18  | 0.30  | 0.09  |
| Lahore     | 4.14  | 5.55  | 1.10  | 0.36  | 1.06  | 2.08  | 1.44  | 15.63 |
| Average    | 1.11  | 1.46  | 1.11  | 0.91  | 1.00  | 1.23  | 0.85  | 3.07  |

Table 8. Cont.

| Cities     | ANN10 | ANN11 | ANN13 | ANN29 | ANN30 | ANN31 | ANN43 | ANN45 |
|------------|-------|-------|-------|-------|-------|-------|-------|-------|
| EMNN       |       |       |       |       |       |       |       |       |
| Peshawar   | 5.28  | 2.03  | 2.26  | 3.70  | 0.14  | 2.86  | 2.07  | 2.39  |
| Bahawalpur | 0.85  | 3.20  | 0.76  | 0.25  | 0.25  | 2.17  | 0.09  | 1.89  |
| Hyderabad  | 0.04  | 0.56  | 0.04  | 0.04  | 0.22  | 0.35  | 0.15  | 0.32  |
| Islamabad  | 0.98  | 2.00  | 1.15  | 1.45  | 0.12  | 0.31  | 1.18  | 2.76  |
| Karachi    | 0.51  | 0.66  | 1.82  | 0.81  | 0.39  | 2.12  | 0.02  | 2.10  |
| Khuzdar    | 0.84  | 1.01  | 0.35  | 0.68  | 0.24  | 1.00  | 0.04  | 1.02  |
| Multan     | 0.49  | 0.97  | 0.76  | 1.08  | 0.82  | 0.33  | 1.92  | 1.00  |
| Quetta     | 0.29  | 0.57  | 0.63  | 1.13  | 1.07  | 0.03  | 1.12  | 2.08  |
| Lahore     | 1.06  | 0.79  | 9.51  | 8.55  | 0.93  | 0.69  | 6.50  | 14.60 |
| Average    | 1.15  | 1.31  | 1.92  | 1.97  | 0.47  | 1.10  | 1.34  | 3.13  |

Figures 5 and 6 compare daily and monthly FFNN GHI to the measured GHI for each site, with the most accurate model name listed in the upper right corner of the location map. Figure 5 shows that, except for sharp peaks, the predicted GHI follows the measured GHI pattern. For the performance evaluation of the CFFN model, we examined eight models, as shown in Table 3. For daily, monthly, and annual GHI prediction, the ANN10, ANN29, and ANN30 performed best. Table 5 indicates that the GHI predicted by all CFFN models was better than satellite-derived GHI. For daily prediction, the average MAPE, RMSE, and MBE values were 8.22%, 0.50 kWh/m<sup>2</sup>/d, and 0.003 kWh/m<sup>2</sup>/d. The average MAPE, RMSE, and MBE for monthly prediction were 3.41%, 0.15 kWh/m<sup>2</sup>/d, and 0.15 kWh/m<sup>2</sup>/d, and 0.86%, 0.18 kWh/m<sup>2</sup>/d and 0.05 kWh/m<sup>2</sup>/d, for annual prediction. The comparisons between CFFN-projected daily and monthly GHI and the measured GHI for each site are shown in Figures 7 and 8, with the most accurate model in the upper right of each location subplot.

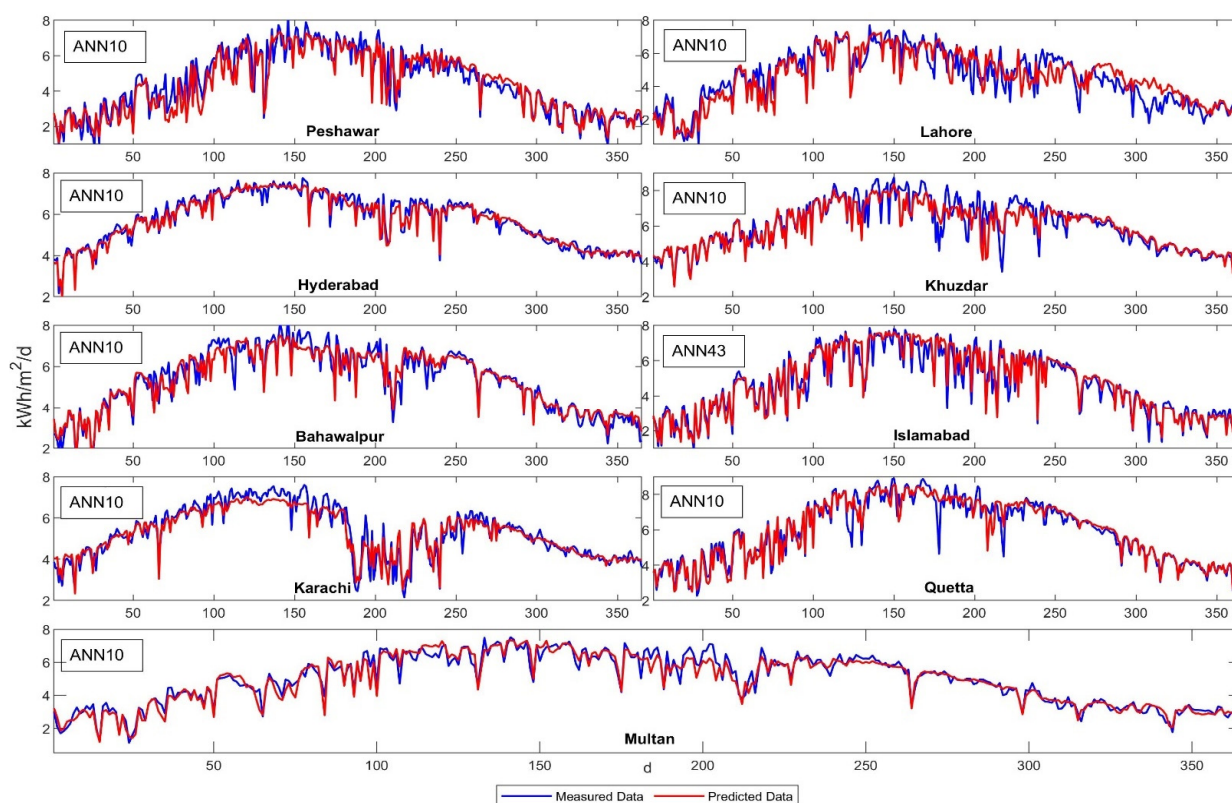


Figure 5. FFNN-predicted GHI vs. measured GHI.

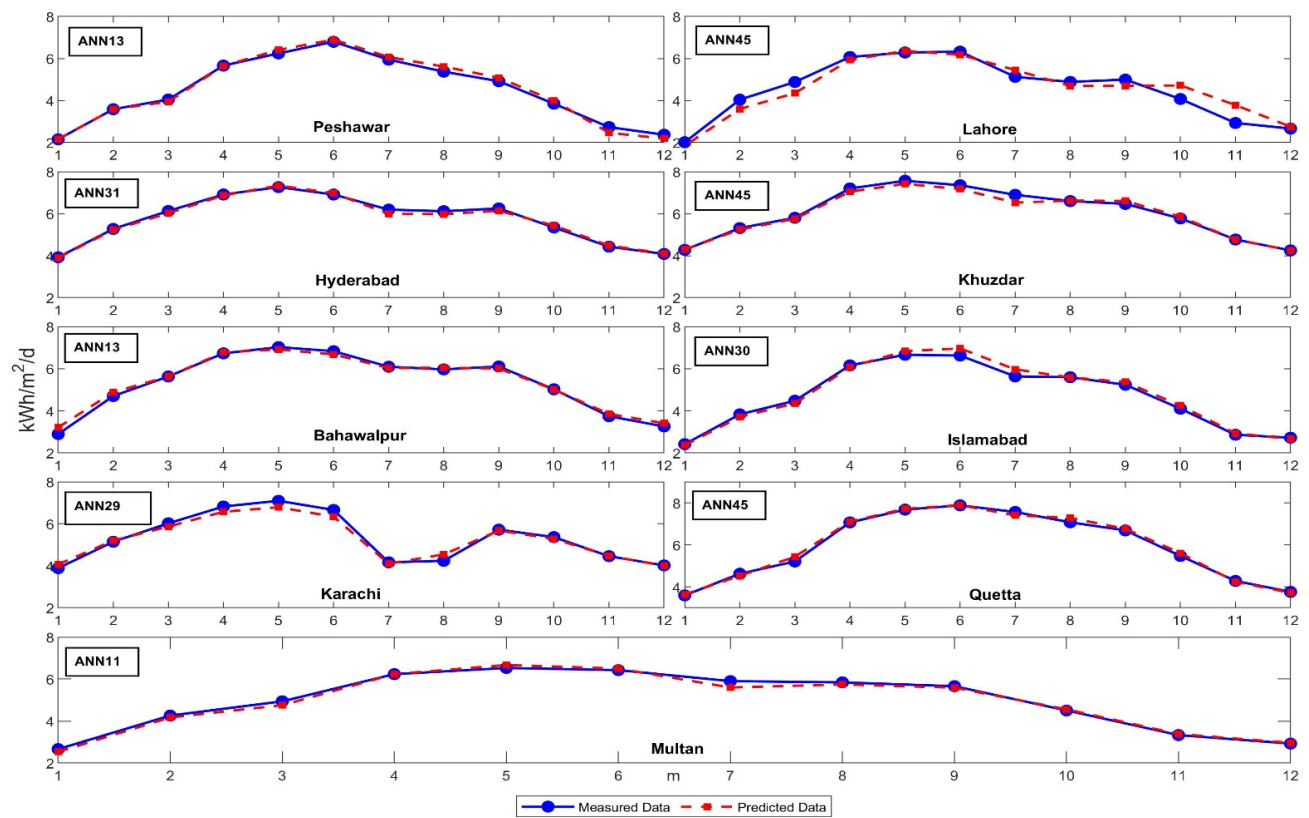


Figure 6. Monthly FFNN-predicted GHI vs. measured GHI.

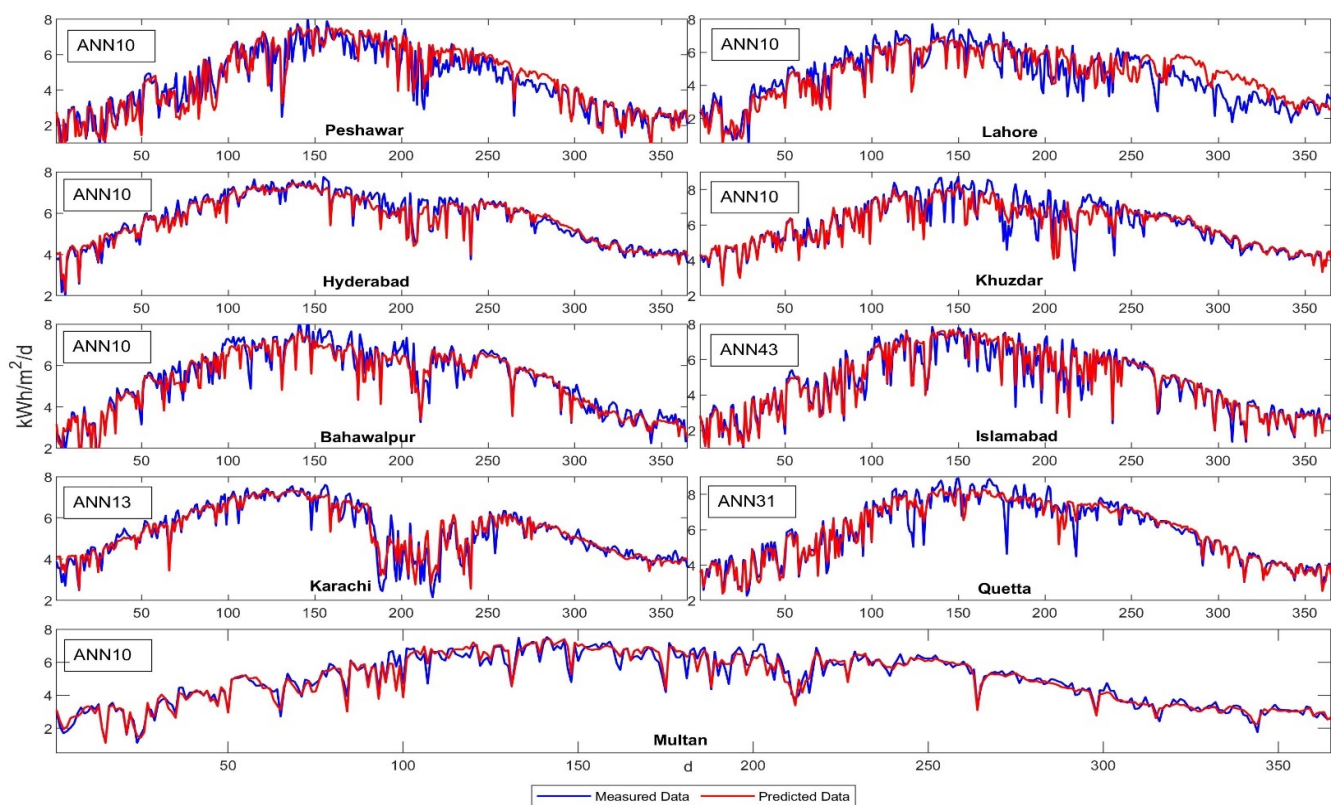


Figure 7. Daily CFNN-predicted GHI vs. measured GHI.

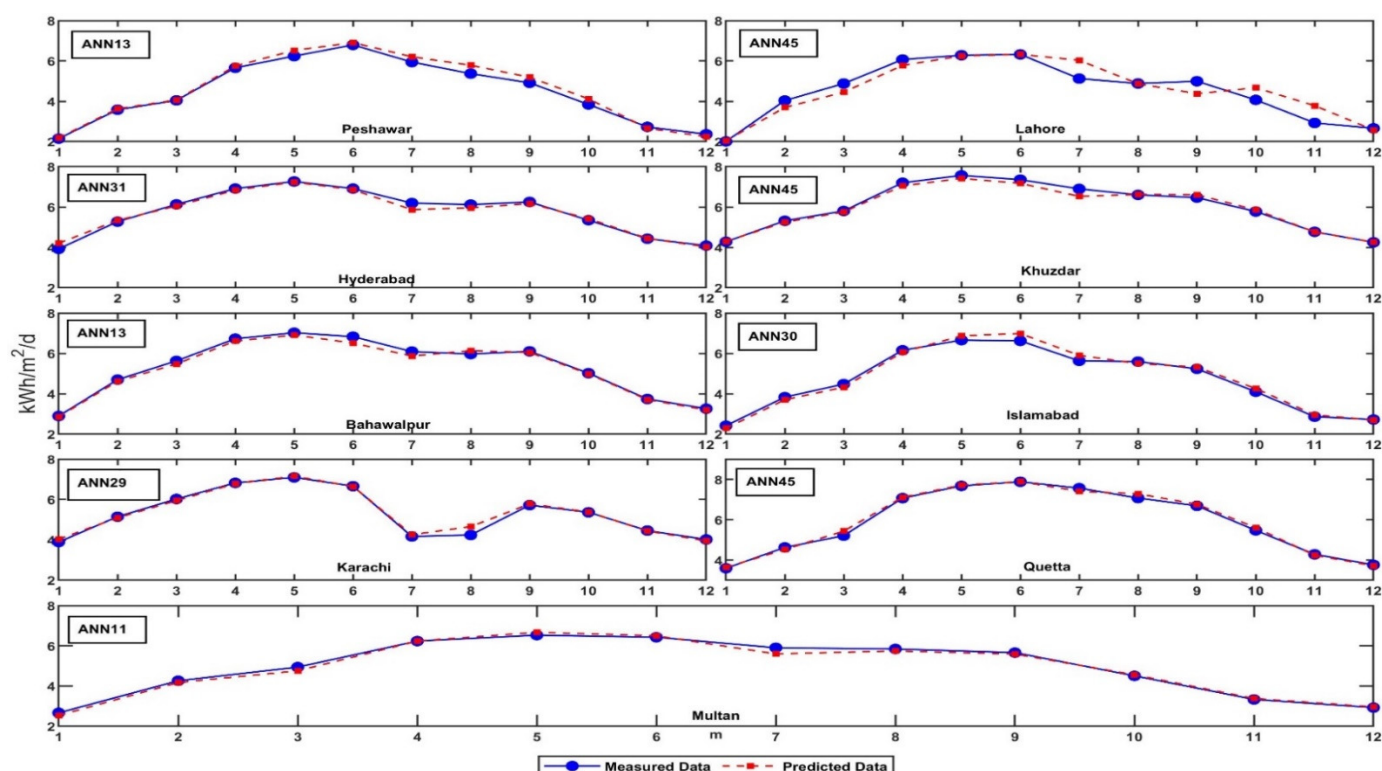


Figure 8. Monthly CFNN-predicted GHI vs. measured GHI.

Compared to the CFFN and FFNN models, the EMNN model had higher MAPE, RMSE, and MBE values for daily GHI prediction, as shown in Table 5. Compared to the CFNN and FFNN models, the EMNN model had the highest accuracy for monthly and annual forecasting. In terms of daily predictions, the best model average MAPE, RMSE, and MBE values were 8.23%, 0.51 kWh/m<sup>2</sup>/d, and 0.05 kWh/m<sup>2</sup>/d. The average MAPE, RMSE, and MBE values in monthly prediction were 3.45%, 0.16 kWh/m<sup>2</sup>/d, and 0.16 kWh/m<sup>2</sup>/d, and 0.47%, 0.18 kWh/m<sup>2</sup>/d, and 0.03 kWh/m<sup>2</sup>/d for annual prediction. The comparisons between EMNN-predicted daily and monthly GHI and measured GHI for each site are shown in Figures 9 and 10, with the correct model in the upper right of each location subplot.

We noticed that RH,  $G_{sat}$ , and Alt were all present in the top eight models. Another point to consider is that if the  $G_{sat}$  error is high, so is the predicted GHI error. GHI, RH, and Alt are the most influential input parameters. The FFNN model had the best predictions among the networks for daily value, and the EMNN model had the worst. The EMNN model had the best predictions for monthly and annual values, and the FFNN model had the worst. Furthermore, it is clear from Tables 4 and 5 that all networks predicted GHI more accurately than the satellite. The FFNN model reduced the daily MAPE, RMSE, and MBE by 25.4%, 0.11 (kWh/m<sup>2</sup>/d), and 0.01 (kWh/m<sup>2</sup>/d), the monthly MAPE, RMSE, and MBE by 47.3%, 0.13 kWh/m<sup>2</sup>/d, and 0.13 kWh/m<sup>2</sup>/d, and the annual MAPE, RMSE, and MBE by 81.14%, 0.12 kWh/m<sup>2</sup>/d, and 0.19 kWh/m<sup>2</sup>/d. The daily MAPE, RMSE, and MBE reduction by the CFFN model was 21.7%, 0.1 kWh/m<sup>2</sup>/d, and 0.017 kWh/m<sup>2</sup>/d, the monthly reduction was 49.5%, 0.13 kWh/m<sup>2</sup>/d, and 0.13 kWh/m<sup>2</sup>/d, while the annual reduction was 83.6%, 0.11 kWh/m<sup>2</sup>/d, 0.18 kWh/m<sup>2</sup>/d. For the EMNN model, it reduced the daily MAPE and RMSE by 21.6% and 0.1 kWh/m<sup>2</sup>/d, while the MBE was increased by 0.03 kWh/m<sup>2</sup>/d. For monthly, the EMNN model performed best and reduced the MAPE, RMSE, and MBE by 50.62%, 0.13 kWh/m<sup>2</sup>/d, and 0.13 kWh/m<sup>2</sup>/d, while for annual values, the reduction was 91.6%, 0.11, and 0.2.

The EMNN model best predicted the monthly and annual GHI predictions, but the EMNN model is more time and resource-consuming. As the number of neurons in the

hidden layer increases by 20, it takes almost 3 to 4 min per simulation, while the FFNN and CFNN models take 1 to 2 s per simulation.

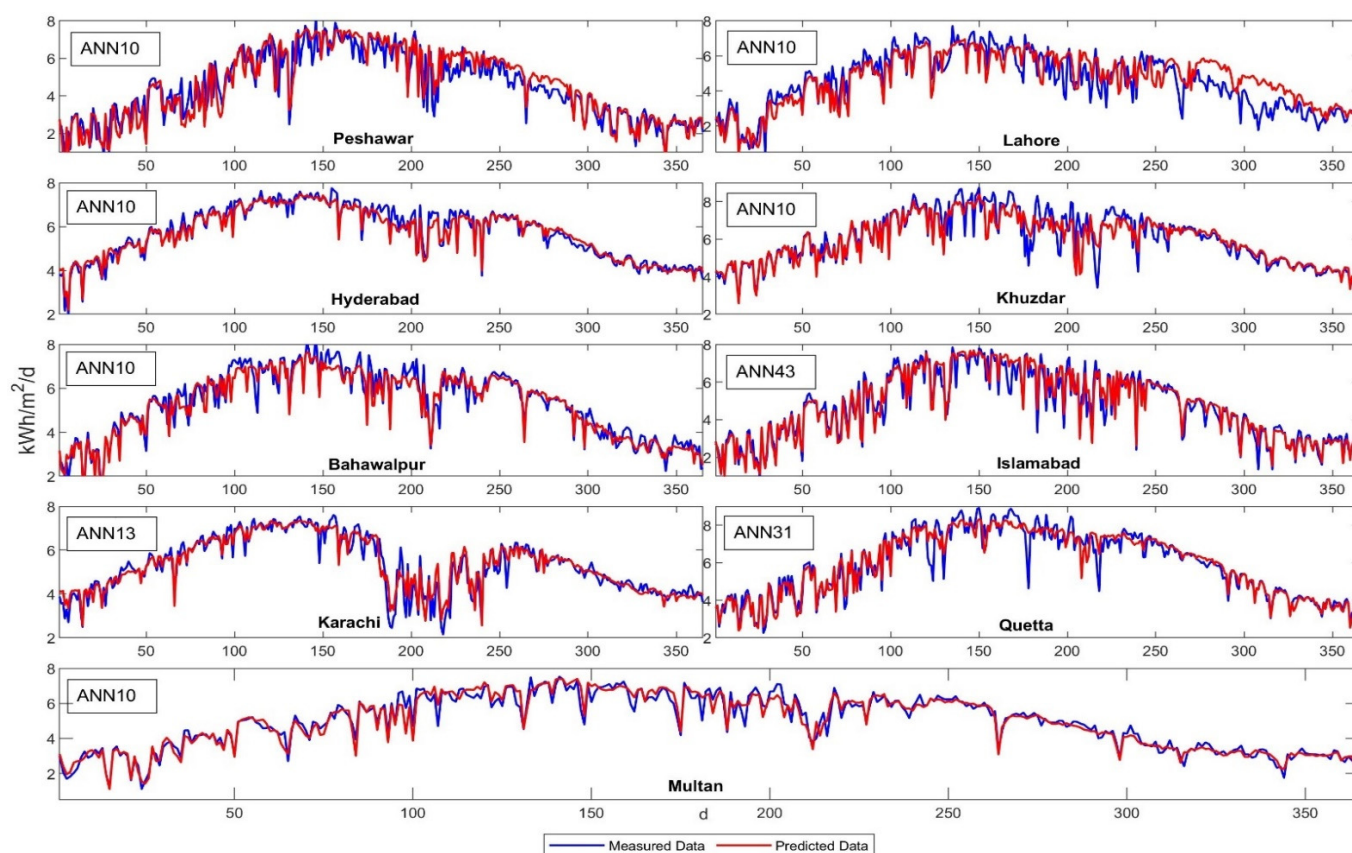


Figure 9. Daily EMNN-predicted GHI vs. measured GHI.

The reason that the FFNN model performed best for daily GHI prediction and worst for yearly GHI prediction is that the FFNN model may demand a large quantity of data for training, and we had 720 datapoints for each city in the daily GHI predictions, totaling 6480 datapoints. While we only have 192 datapoints in the monthly predictions, this may be insufficient for FFNNs to develop an accurate correlation between inputs and targets. While the results for the EMNN and CFNN models improved when the amount of data decreased. The large volume of data may lead the CFNN and EMNN models to become confused. Therefore, the EMNN and CFNN models outperform the FFNN model for monthly and annual GHI prediction.

Our results are more accurate than Agbulut, Gurul, and Bicen [19], as their ANN model MAPE ranged from 15.92% to 22.56%, while our model MAPE ranges from 3.37% to 16.49%. This increase in accuracy is due to testing different combinations of inputs and a wide range of neurons in hidden layers. This is because they just used five input parameters, including  $T_{\max}$ ,  $T_{\min}$ , cloud cover (CC), extraterrestrial radiation, and SH, and did not test the RH and other important parameters for training ANN.

Our results agree with Kuhe, Achirgenda, and Agada [16,25], as the MAPE of their FFNN model was 5.67%, and our models' MAPes, for most of the cities, range from 3.37% to 8.08%, except for Peshawar and Lahore. The results of Braham Belmahdi, Mohamed Louzazni, and Abdelmajid El Bouardi [17] were more accurate than ours, as their model's MAPE was 1.80%. ANN results are dependent on data correlation. Due to the higher inaccuracy of satellite data in these cities, the MAPE in Lahore and Peshawar was substantially higher than in other cities. As a result, it is probable that the data they were working with contained less nonlinearities. In addition, the Tegenu Argaw Woldegiyorgis et al. [18] results, in terms of RMSE, are consistent with the findings of our study, as their

ANN model RMSE value was  $0.331 \text{ kWh/m}^2/\text{d}$ , whereas our model RMSE values vary from  $0.24 \text{ kWh/m}^2/\text{d}$  to  $0.77 \text{ kWh/m}^2/\text{d}$ .

The results also reveal that the ANN models' predictions are more precise in places where satellite data are accurate. For example, the satellite GHI MAPE is 25.31 percent and 18.32 percent in Lahore and Peshawar. The MAPE of the models for Lahore and Peshawar in Tables 5 and 6 are also in the double digits. The satellite GHI MAPE is the lowest in Hyderabad; similarly, ANN model MAPEs are the lowest of all the cities. This explains how the accuracy of satellite data affects the accuracy of the ANN model. If someone is planning to use satellite data, comparing satellite data GHI with ground-measured GHI can predict the range of ANN accuracy. The models were put to the test in various climate zones. It was discovered that climatic zones have an impact on the accuracy of satellite data inputs, as well as on the output of ANN models. As a result, it is recommended that, before applying these models to any place on the planet, satellite data from training sets must be checked for accuracy, which will help the researcher to predict the pre-simulation accuracy of models against the ground data. Whether the satellite accuracy is higher or lower, ANN models will decrease the error rate of satellite data.

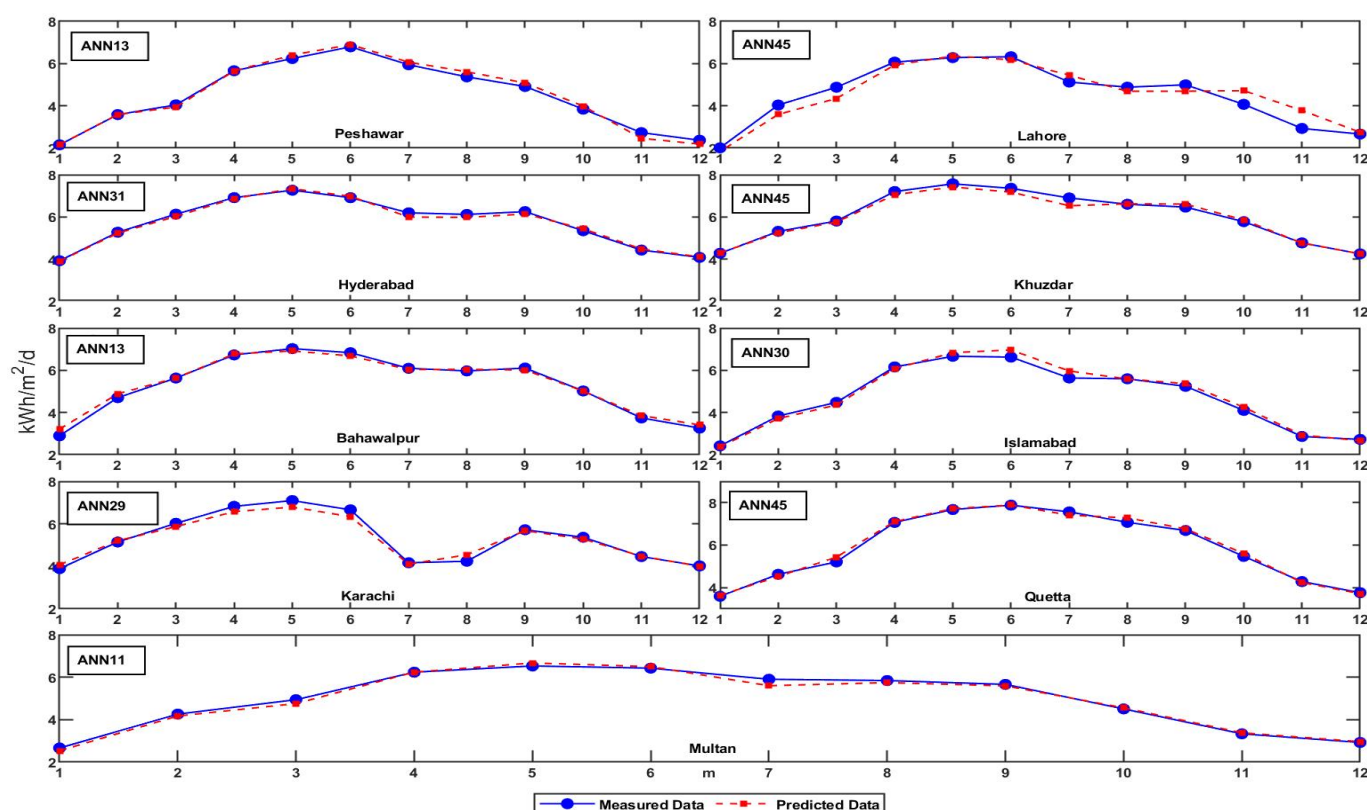


Figure 10. Monthly EMNN-predicted GHI vs. measured GHI.

The models can be utilized in any place with or without a meteorological station, according to the discussion. Because the model can be trained on data from nearby stations before being applied to satellite data. These models have overcome the limitations of previous models, which could only predict GHI for locations with meteorological stations. Reliance on satellite GHI accuracy is the weakness of these models, as when satellite data GHI accuracy decreases, model accuracy decreases as well.

We not only created an ANN model for GHI prediction in this study, but we also discovered the satellite data error rate for nine cities across various climates, which is the primary cause of incorrect solar project sizing. The model will aid solar project engineers in conducting accurate feasibility studies, resulting in the long-term functioning of solar systems and more people agreeing to switch to solar energy, resulting in the sustainable de-

velopment of solar energy systems. This will assist the country in reducing its dependency on fossil fuels and its associated import expenditure.

## 6. Conclusions

This research investigated how well artificial neural networks predict daily, monthly, and annual global horizontal irradiation in Pakistan. The FFNN, CFFN, and EMNN models were tested for nine cities in Pakistan in order to construct a model that can more accurately predict than satellite data. The FFNN model outperformed the CFFN and EMNN models for daily GHI prediction, while EMNN's findings were more accurate for monthly and annual predictions. For daily ANN models with the inputs of latitude, longitude, altitude, day, maximum temperature, minimum temperature, relative humidity, absolute humidity, clearness index, and precipitation, satellite global horizontal irradiation was predicted with the highest accuracy for all three types of networks. A monthly GHI prediction model with the inputs of latitude, longitude, altitude, month, day, precipitation, relative humidity, satellite global horizontal irradiation should be used with the EMNN model, while an EMNN with the inputs of latitude, longitude, altitude, day, precipitation, relative humidity, satellite global horizontal irradiation should be used for annual GHI predictions. The predictions from all three networks were more accurate than the available satellite data.

As in this research, inputs from satellite data were used, which covered all areas in the country. The models were trained on data from diverse locations across the different climatic and geographical zones of Pakistan. As a result, regardless of whether a meteorological station is available, all three networks can be used in any area in the country.

The model's accuracy depends primarily on the accuracy of satellite data, rather than geographical location or climate, and it produces more accurate results than satellite data, enabling it to be used in any location around the world. In any case, the model will decrease satellite data error by 25% to 50%.

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## References

1. El-Kenawy, E.-S.M.; Mirjalili, S.; Ghoneim, S.S.; Eid, M.M.; El-Said, M.; Khan, Z.S.; Ibrahim, A. Advanced Ensemble Model for Solar Radiation Forecasting Using Sine Cosine Algorithm and Newton's Laws. *IEEE Access* **2021**, *9*, 115750–115765. [[CrossRef](#)]
2. Zhang, Z.; Wang, C.; Peng, X.; Qin, H.; Lv, H.; Fu, J.; Wang, H. Solar Radiation Intensity Probabilistic Forecasting Based on K-Means Time Series Clustering and Gaussian Process Regression. *IEEE Access* **2021**, *9*, 89079–89092. [[CrossRef](#)]

3. Braik, A.H.; Sopian, K.; Jarimi, H.; Fazlizan, A.; Ibrahim, A.; Abd Hamid, A.S. Geospatial Analysis of Photovoltaic Plants Potential in Palestine Using a GIS-AHP-Based Technique. *Int. J. Renew. Energy Res. (IJRER)* **2022**, *12*, 479–488.
4. Khatib, T.; Mohamed, A.; Sopian, K.; Mahmoud, M. Assessment of artificial neural networks for hourly solar radiation prediction. *Int. J. Photoenergy* **2012**, *2012*, 1–7. [\[CrossRef\]](#)
5. Hamid, A.; Ibrahim, A.; Assadeg, J.; Ahmad, E.; Sopian, K. Techno-economic Analysis of a Hybrid Solar Dryer with a Vacuum Tube Collector for Hibiscus Cannabinus L Fiber. *Int. J. Renew. Energy Res.* **2020**, *10*, 1609–1613.
6. Patchali, T.E.; Oyewola, O.M.; Ajide, O.O.; Matthew, O.J.; Salau, T.A.; Adaramola, M.S. Assessment of global solar radiation estimates across different regions of Togo, West Africa. *Meteorol. Atmos. Phys.* **2022**, *134*, 1–15. [\[CrossRef\]](#)
7. Oyewola, O.M.; Patchali, T.E.; Ajide, O.O.; Singh, S.; Matthew, O.J. Global solar radiation predictions in Fiji Islands based on empirical models. *Alex. Eng. J.* **2022**, *61*, 8555–8571. [\[CrossRef\]](#)
8. Mustafa, J.; Alqaed, S.; Almeshmadi, F.A.; Jamil, B. Development and comparison of parametric models to predict global solar radiation: A case study for the southern region of Saudi Arabia. *J. Therm. Anal. Calorim.* **2022**, 1–31. [\[CrossRef\]](#)
9. Abd Hamid, A.S.; Razali, H. Solar Car: Brief Review and Challenges. *Borneo Sci. J.* **2019**, *40*, 27–37.
10. Besharat, F.; Dehghan, A.A.; Faghih, A.R. Empirical models for estimating global solar radiation: A review and case study. *Renew. Sustain. Energy Rev.* **2013**, *21*, 798–821. [\[CrossRef\]](#)
11. Wang, L.; Kisi, O.; Zounemat-Kermani, M.; Salazar, G.A.; Zhu, Z.; Gong, W. Solar radiation prediction using different techniques: Model evaluation and comparison. *Renew. Sustain. Energy Rev.* **2016**, *61*, 384–397. [\[CrossRef\]](#)
12. Qin, W.; Wang, L.; Lin, A.; Zhang, M.; Xia, X.; Hu, B.; Niu, Z. Comparison of deterministic and data-driven models for solar radiation estimation in China. *Renew. Sustain. Energy Rev.* **2018**, *81*, 579–594. [\[CrossRef\]](#)
13. Kaba, K.; Sarıgül, M.; Avcı, M.; Kandırmaz, H.M. Estimation of daily global solar radiation using deep learning model. *Energy* **2018**, *162*, 126–135. [\[CrossRef\]](#)
14. Aljanad, A.; Tan, N.M.; Agelidis, V.G.; Shareef, H. Neural Network Approach for Global Solar Irradiance Prediction at Extremely Short-Time-Intervals Using Particle Swarm Optimization Algorithm. *Energies* **2021**, *14*, 1213. [\[CrossRef\]](#)
15. Khatib, T.; Mohamed, A.; Sopian, K.; Mahmoud, M. Modeling of solar energy for Malaysia using artificial neural networks. In Proceedings of the 11th WSEAS/IASME International Conference on Electric Power Systems, High Voltages, Electric Machines, Penang, Malaysia, 3–5 October 2011; pp. 486–489. Available online: [https://d1wqtxts1xzle7.cloudfront.net/38267684/ICOPOW-13-with-cover-page-v2.pdf?Expires=1656249562&Signature=R0vhdVyVcYRbBD8AWd2xZ4nSK7JghLuQuZVaqnmlasldhR3dKS0gWeXUblXVHRF0uoyQr32QZw3bkT4Utk3LZfpoTLkLyZyBlt51Q3Cv-QQZ4Fa0QCfzVswicirgkAEMRob3-7st~{}SJ3m08GncSamLIZNFwvmWBBzTkjpan0LQWCld48woBq4DbS-qp90Dmc1nnsFcInaIajLCUE-DvHSXw1o5U6kK3-oZE1oolpHi5DbRr3Zw7zH1m67a1s8ScpI7U-QUdGXR8ut4WpiJhd3NaXoxSuq5N7grXIAP~{}E8RC5-Mv4OQoIsE8ShuB0YvcKEqhcgUNC51kWo4Y5Swj-tA\\_\\_&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA](https://d1wqtxts1xzle7.cloudfront.net/38267684/ICOPOW-13-with-cover-page-v2.pdf?Expires=1656249562&Signature=R0vhdVyVcYRbBD8AWd2xZ4nSK7JghLuQuZVaqnmlasldhR3dKS0gWeXUblXVHRF0uoyQr32QZw3bkT4Utk3LZfpoTLkLyZyBlt51Q3Cv-QQZ4Fa0QCfzVswicirgkAEMRob3-7st~{}SJ3m08GncSamLIZNFwvmWBBzTkjpan0LQWCld48woBq4DbS-qp90Dmc1nnsFcInaIajLCUE-DvHSXw1o5U6kK3-oZE1oolpHi5DbRr3Zw7zH1m67a1s8ScpI7U-QUdGXR8ut4WpiJhd3NaXoxSuq5N7grXIAP~{}E8RC5-Mv4OQoIsE8ShuB0YvcKEqhcgUNC51kWo4Y5Swj-tA__&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA) (accessed on 24 May 2022).
16. Kuhe, A.; Achirgbenda, V.T.; Agada, M. Global solar radiation prediction for Makurdi, Nigeria, using neural networks ensemble. *Energy Sources Part A Recovery Util. Environ. Eff.* **2021**, *43*, 1373–1385. [\[CrossRef\]](#)
17. Belmahdi, B.; Louzazni, M.; El Bouardi, A. Comparative optimization of global solar radiation forecasting using machine learning and time series models. *Environ. Sci. Pollut. Res.* **2022**, *29*, 14871–14888. [\[CrossRef\]](#)
18. Woldegiyorgis, T.A.; Admasu, A.; Benti, N.E.; Asfaw, A.A. A Comparative Evaluation of Artificial Neural Network and Sunshine Based models in prediction of Daily Global Solar Radiation of Lalibela, Ethiopia. *Cogent Eng.* **2022**, *9*, 1996871. [\[CrossRef\]](#)
19. Ağbulut, Ü.; Gürel, A.E.; Biçen, Y. Prediction of daily global solar radiation using different machine learning algorithms: Evaluation and comparison. *Renew. Sustain. Energy Rev.* **2021**, *135*, 110114. [\[CrossRef\]](#)
20. Mahdi, B.H.; Yousif, K.M.; Dosky, L.M. Using Artificial Neural Networks to Predict Solar Radiation for Duhok City, Iraq. In Proceedings of the 2020 International Conference on Computer Science and Software Engineering (CSASE), Duhok, Iraq, 16–18 April 2020; pp. 1–6.
21. Ghimire, S.; Deo, R.C.; Downs, N.J.; Raj, N. Global solar radiation prediction by ANN integrated with European Centre for medium range weather forecast fields in solar rich cities of Queensland Australia. *J. Clean. Prod.* **2019**, *216*, 288–310. [\[CrossRef\]](#)
22. Sözen, A.; Arcaklioğlu, E.; Özalp, M.; Kanit, E.G. Use of artificial neural networks for mapping of solar potential in Turkey. *Appl. Energy* **2004**, *77*, 273–286. [\[CrossRef\]](#)
23. Şenkal, O.; Kuleli, T. Estimation of solar radiation over Turkey using artificial neural network and satellite data. *Appl. Energy* **2009**, *86*, 1222–1228. [\[CrossRef\]](#)
24. Khatib, T.; Mohamed, A.; Mahmoud, M.; Sopian, K. Modeling of daily solar energy on a horizontal surface for five main sites in Malaysia. *Int. J. Green Energy* **2011**, *8*, 795–819. [\[CrossRef\]](#)
25. Tahir, Z.; Asim, M. Surface measured solar radiation data and solar energy resource assessment of Pakistan: A review. *Renew. Sustain. Energy Rev.* **2018**, *81*, 2839–2861. [\[CrossRef\]](#)
26. Azhar, M.; Blanc, P.; Asim, M.; Imran, S.; Hayat, N.; Shahid, H.; Ali, H. The evaluation of reanalysis and analysis products of solar radiation for Sindh province, Pakistan. *Renew. Energy* **2020**, *145*, 347–362.
27. Sukhera, M.; Pasha, M. Solar radiation maps for Pakistan. *Sol. Wind. Technol.* **1987**, *4*, 229–238. [\[CrossRef\]](#)
28. Adnan, S.; Hayat Khan, A.; Haider, S.; Mahmood, R. Solar energy potential in Pakistan. *J. Renew. Sustain. Energy* **2012**, *4*, 032701. [\[CrossRef\]](#)
29. Ghaffar, M.A. The energy supply situation in the rural sector of Pakistan and the potential of renewable energy technologies. *Renew. Energy* **1995**, *6*, 941–976. [\[CrossRef\]](#)

30. Chaudhary, Q.-U.-Z. A procedure to obtain global solar radiation maps from sunshine duration for Pakistan. *Sol. Wind. Technol. (UK)* **1990**, *7*, 245–253. [[CrossRef](#)]
31. Raja, I.; Twidell, J. Distribution of global insolation over Pakistan. *Sol. Energy* **1989**, *43*, 355–357. [[CrossRef](#)]
32. Garud, K.S.; Jayaraj, S.; Lee, M.Y. A review on modeling of solar photovoltaic systems using artificial neural networks, fuzzy logic, genetic algorithm and hybrid models. *Int. J. Energy Res.* **2021**, *45*, 6–35. [[CrossRef](#)]
33. Hoang, A.T.; Nižetić, S.; Ong, H.C.; Tarelko, W.; Le, T.H.; Chau, M.Q.; Nguyen, X.P. A review on application of artificial neural network (ANN) for performance and emission characteristics of diesel engine fueled with biodiesel-based fuels. *Sustain. Energy Technol. Assess.* **2021**, *47*, 101416.
34. Toghraie, D.; Sina, N.; Jolfaei, N.A.; Hajian, M.; Afrand, M. Designing an Artificial Neural Network (ANN) to predict the viscosity of Silver/Ethylene glycol nanofluid at different temperatures and volume fraction of nanoparticles. *Phys. A Stat. Mech. Its Appl.* **2019**, *534*, 122142. [[CrossRef](#)]
35. Khatib, T. A novel approach for solar radiation prediction using artificial neural networks. *Energy Sources Part A Recovery Util. Environ. Eff.* **2015**, *37*, 2429–2436. [[CrossRef](#)]
36. Milad, A.; Adwan, I.; Majeed, S.A.; Yusoff, N.I.M.; Al-Ansari, N.; Yaseen, Z.M. Emerging technologies of deep learning models development for pavement temperature prediction. *IEEE Access* **2021**, *9*, 23840–23849. [[CrossRef](#)]
37. Cheng, Y.-C.; Qi, W.-M.; Cai, W.-Y. Dynamic properties of Elman and modified Elman neural network. In Proceedings of the International Conference on Machine Learning and Cybernetics, Beijing, China, 4–5 November 2002; pp. 637–640.
38. Sparks, A.H. nasapower: A NASA POWER global meteorology, surface solar energy and climatology data client for R. *J. Open Source Softw.* **2018**, *3*, 1035. [[CrossRef](#)]
39. Mubiru, J. Predicting total solar irradiation values using artificial neural networks. *Renew. Energy* **2008**, *33*, 2329–2332. [[CrossRef](#)]
40. Lewis, C.D. *Industrial and Business Forecasting Methods: A Practical Guide to Exponential Smoothing and Curve Fitting*; Butterworth-Heinemann: Oxford, UK, 1982.